Year 10 Foundation

REVISION PACK

Algebra 2

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Algebra 2 · Indices in algebra (Lesson 7.1)

Q	Working	Answer
simplifying with index laws		
1.1	**(a)** x^{2+7} **(b)** y^{10-4} **(c)** $5^2 = 25,\ 2 \times 2 = 4$ **(d)** $8 \div 2 = 4,\ m^{7-3},\ n^{3-1}$	**(a)** x^9 **(b)** y^6 **(c)** $25x^4$ **(d)** $4m^4n^2$
1.2	**(a)** a^{5+3} **(b)** b^{7-2} **(c)** $2^3 = 8,\ 4 \times 3 = 12$ **(d)** $15 \div 3 = 5,\ x^{5-2},\ y^{4-1}$	**(a)** a^8 **(b)** b^5 **(c)** $8a^{12}$ **(d)** $5x^3y^3$
1.3	**(a)** p^{4+6} **(b)** q^{9-3} **(c)** $4^2 = 16,\ 5 \times 2 = 10$ **(d)** $12 \div 4 = 3,\ p^{9-4},\ q^{2-2}$	**(a)** p^{10} **(b)** q^6 **(c)** $16y^{10}$ **(d)** $3p^5$
1.4	**(a)** m^{3+8} **(b)** n^{12-5} **(c)** $3^3 = 27,\ 2 \times 3 = 6$ **(d)** $20 \div 5 = 4,\ a^{6-2},\ b^{4-3}$	**(a)** m^{11} **(b)** n^7 **(c)** $27x^6$ **(d)** $4a^4b$
1.5	**(a)** x^{6+4} **(b)** y^{11-2} **(c)** $6^2 = 36,\ 3 \times 2 = 6$ **(d)** $18 \div 9 = 2,\ m^{5-4},\ n^{3-1}$	**(a)** x^{10} **(b)** y^9 **(c)** $36b^6$ **(d)** $2mn^2$

1.6	**(a)** t^{7+1} **(b)** k^{15-5} **(c)** $2^4 = 16,\ 5 \times 4 = 20$ **(d)** $14 \div 7 = 2,\ x^{8-3},\ y^{5-2}$	**(a)** t^8 **(b)** k^{10} **(c)** $16c^{20}$ **(d)** $2x^5y^3$